

THE FRONTIER

Low-Level Requirements Document -- Version 0.5.1

Traces to: HLR v0.5.0 | GDD v8 Target: Godot 4.x | GDScript | JSON data March 2026

| T = Testable | U = Untestable (subjective/emergent)

1. Core Application

LLR	Traces To	Requirement	T/U
LCORE-001	CORE-001	The project shall use Godot 4.x as its engine.	T
LCORE-002	CORE-001	The main scene shall be a single main.tscn that loads a WorldEnvironment, the player scene, and a map scene.	T
LCORE-003	CORE-001	The map scene shall be a separate PackedScene loaded at runtime based on a config JSON file.	T
LCORE-004	CORE-002	A debug overlay toggled via a key shall display current FPS, draw calls, and node count.	T
LCORE-005	CORE-002	The project shall set target_fps in Project Settings.	T
LCORE-006	CORE-003	An InputManager autoload shall load keybindings from user://input_config.json.	T
LCORE-007	CORE-003	Default keybindings for keyboard/mouse and gamepad shall ship in res://data/default_input.json.	T
LCORE-008	CORE-004	A SaveManager autoload shall serialize world state to user://saves/<slot>/save.json.	T
LCORE-009	CORE-004	SaveManager shall auto-save when it receives the sleep_complete signal.	T
LCORE-010	CORE-004	SaveManager shall support manual save via the pause menu.	T

LLR	Traces To	Requirement	T/U
LCORE-011	CORE-004	SaveManager shall support N save slots.	T
LCORE-012	CORE-004	Slot metadata (name, date, playtime, settlement count) shall be stored in user://saves/slots.json.	T
LCORE-013	CORE-005	A DataLoader autoload shall parse all JSON files from res://data/ at startup.	T
LCORE-014	CORE-005	DataLoader shall expose typed dictionaries (monsters, items, areas, tiers, disciplines, etc.) accessible by all systems.	T

2. Data Layer

2.1 JSON Files -- Existence and Parse

LLR	Traces To	Requirement	T/U
LDATA-001	DATA-001	res://data/monsters.json shall exist and be parseable by DataLoader.	T
LDATA-002	DATA-001	res://data/spawn_tables.json shall exist and be parseable by DataLoader.	T
LDATA-003	DATA-001	res://data/resources.json shall exist and be parseable by DataLoader.	T
LDATA-004	DATA-001	res://data/areas.json shall exist and be parseable by DataLoader.	T
LDATA-005	DATA-001	res://data/settlement_tiers.json shall exist and be parseable by DataLoader.	T
LDATA-006	DATA-001	res://data/trade_weights.json shall exist and be parseable by DataLoader.	T
LDATA-007	DATA-001	res://data/export_thresholds.json shall exist and be parseable by DataLoader.	T

LLR	Traces To	Requirement	T/U
LDATA-008	DATA-001	res://data/disciplines.json shall exist and be parseable by DataLoader.	T
LDATA-009	DATA-001	res://data/reagents.json shall exist and be parseable by DataLoader.	T
LDATA-010	DATA-001	res://data/runes.json shall exist and be parseable by DataLoader.	T
LDATA-011	DATA-001	res://data/items.json shall exist and be parseable by DataLoader.	T
LDATA-012	DATA-001	res://data/injuries.json shall exist and be parseable by DataLoader.	T
LDATA-013	DATA-001	res://data/weapons.json shall exist and be parseable by DataLoader.	T
LDATA-014	DATA-001	res://data/seasons.json shall exist and be parseable by DataLoader.	T
LDATA-015	DATA-001	res://data/survival.json shall exist and be parseable by DataLoader.	T
LDATA-016	DATA-001	res://data/recipes.json shall exist and be parseable by DataLoader.	T
LDATA-017	DATA-001	res://data/manufactured_goods.json shall exist and be parseable by DataLoader.	T
LDATA-018	DATA-001	res://data/hirelings.json shall exist and be parseable by DataLoader.	T
LDATA-019	DATA-001	res://data/roads.json shall exist and be parseable by DataLoader.	T
LDATA-020	DATA-001	res://data/landmarks.json shall exist and be parseable by DataLoader.	T

2.2 Key Schema Requirements

LLR	Traces To	Requirement	T/U
LDATA-030	DATA-001	Each monster entry shall include: id, name, model_path, hp, damage, speed, behavior_type, loot_table, fire_reaction, light_reaction.	T
LDATA-031	DATA-001	Each area entry shall include: id, region_id, hex_coords{q,r}, neighbors[], resource_tags[], spawn_table_id, difficulty, tags[], edge_features[], settlement_site{}, precursor_site_index.	T
LDATA-032	DATA-001	Each area edge_feature entry shall include: edge_index (0-5) and type ("river", "cliff", "ford", "rapids").	T
LDATA-033	DATA-001	Each settlement_tiers entry shall include: tier, name, score_threshold, required_resources[], suppression_pct, safe_zone_radius, consumption_rate, exploitation{own_pct, adjacent_pct, ring2_pct}, services[], road_budget.	T
LDATA-034	DATA-001	Each rune entry shall include: id, name, effects{}, slot_types[], rarity, poisson_mean, poisson_min, poisson_max.	T
LDATA-035	DATA-001	Each item entry shall include: id, name, weight, type, durability, value, rune_slots, noise_level, spoil_timer, tags[].	T
LDATA-036	DATA-001	Waterskin items shall have a weight_empty field separate from weight.	T
LDATA-037	DATA-001	Each injury entry shall include: id, name, debuffs{}, causes[], probability_by_source{}, treatment_tier.	T
LDATA-038	DATA-001	Each weapon entry shall include: id, name, type, damage, speed, stamina_cost, weight, durability, rune_slots, noise, weapon_category.	T
LDATA-039	DATA-001	Each discipline entry shall include: id, name, abilities[], xp_triggers[].	T
LDATA-040	DATA-001	Each reagent entry shall include: id, name, spell_id, weight, tier, source, buy_price.	T
LDATA-041	DATA-001	disciplines.json shall contain exactly 9 disciplines and 61 total abilities.	T

LLR	Traces To	Requirement	T/U
LDATA-042	DATA-001	injuries.json shall contain exactly 7 injury types.	T

3. World and Map

3.1 Hex Grid

LLR	Traces To	Requirement	T/U
LMAP-001	MAP-002	A HexGrid autoload shall convert axial hex coordinates (q,r) to world position.	T
LMAP-002	MAP-002	HexGrid shall convert world position to axial hex coordinates.	T
LMAP-003	MAP-002	HexGrid shall provide neighbor lookup for any hex (returns up to 6 neighbors).	T
LMAP-004	MAP-002	HexGrid shall calculate hex distance between two hexes.	T
LMAP-005	MAP-002	HexGrid shall provide ring queries (all hexes at distance N from a center).	T
LMAP-006	MAP-002	HexGrid shall identify shared edges between adjacent hexes by edge index (0-5).	T

3.2 Area Management

LLR	Traces To	Requirement	T/U
LMAP-010	MAP-003	An AreaManager autoload shall load areas.json at startup.	T
LMAP-011	MAP-003	AreaManager shall maintain runtime state per area: discovered_resources[], settlement_id, suppression_pct, map_revealed.	T

LLR	Traces To	Requirement	T/U
LMAP-012	MAP-003	AreaManager.get_area(pos:Vector3) shall return the area_id for any world position via HexGrid lookup.	T
LMAP-013	MAP-004	River edges shall be stored as edge_features in areas.json, shared between adjacent hexes.	T
LMAP-014	MAP-007	Cliff edges shall be stored as edge_features in areas.json.	T
LMAP-015	MAP-004	AreaManager.is_edge_passable(area_a, area_b) shall return false if the shared edge has a cliff.	T
LMAP-016	MAP-006	AreaManager.is_edge_passable() shall return false if the shared edge has a river and neither area has a Ford tag and neither has a Town+ settlement.	T
LMAP-017	MAP-006	AreaManager.is_edge_passable() shall return true for a river edge if either area has a Ford tag.	T
LMAP-018	MAP-006	AreaManager.is_edge_passable() shall return true for a river edge if the area has a Town+ settlement (bridge).	T

3.3 Time

LLR	Traces To	Requirement	T/U
LMAP-020	MAP-012	A TimeManager autoloader shall track time_of_day as float 0.0-24.0.	T
LMAP-021	MAP-012	30 real-time minutes shall equal 24.0 in-game hours.	T
LMAP-022	MAP-012	The time-to-real-time ratio shall be configurable in JSON.	T
LMAP-023	MAP-012	TimeManager shall drive DirectionalLight rotation based on time_of_day.	T
LMAP-024	MAP-013	TimeManager shall calculate daylight hours by interpolating between season daylight values.	T

LLR	Traces To	Requirement	T/U
LMAP-025	MAP-013	TimeManager shall emit dawn and dusk signals.	T

3.4 Seasons and Weather

LLR	Traces To	Requirement	T/U
LMAP-030	MAP-014	A SeasonManager autoload shall track season_index (0-3) and day_count.	T
LMAP-031	MAP-014	SeasonManager shall advance season when day_count exceeds the current season's duration from seasons.json.	T
LMAP-032	MAP-014	SeasonManager shall emit a season_changed signal on season advance.	T
LMAP-033	MAP-014	Days per season shall be configurable in seasons.json.	T
LMAP-034	MAP-015	A WeatherManager shall roll against the current season's weather_table on each new day.	T
LMAP-035	MAP-015	Active weather shall modify fog density.	T
LMAP-036	MAP-015	Active weather shall toggle rain/snow particle effects.	T
LMAP-037	MAP-015	Active weather shall modify temperature.	T
LMAP-038	MAP-015	Rain weather shall increase effective cold level.	T

3.5 Precursors and Landmarks

LLR	Traces To	Requirement	T/U
LMAP-040	MAP-016	A PrecursorRandomizer shall run on new game creation.	T
LMAP-041	MAP-016	PrecursorRandomizer shall use a seeded RandomNumberGenerator. The seed shall be stored in the save file.	T
LMAP-042	MAP-016	PrecursorRandomizer shall assign at least 18 sites as Places of Power (minimum 2 per discipline).	T
LMAP-043	MAP-016	Remaining precursor sites shall be assigned ruin types (shelter, lore, materials, navigation).	T
LMAP-044	MAP-016	Precursor assignments shall be stored in the save as Dict<site_index, assignment>.	T
LMAP-045	MAP-017	A LandmarkManager shall load landmark definitions from map data.	T
LMAP-046	MAP-017	On player first entry to a landmark trigger (Area3D), a naming prompt shall appear.	T
LMAP-047	MAP-017	Player-assigned landmark names shall persist in the save file.	T
LMAP-048	MAP-017	Named landmarks shall appear on maps in the MapViewer.	T
LMAP-049	MAP-017	From any point above a configurable elevation threshold, at least 3 landmark meshes shall be visible.	U

3.6 Map Scene

LLR	Traces To	Requirement	T/U
LMAP-050	MAP-001	The map shall be a single Godot scene containing terrain meshes, biome materials, water bodies, and vegetation.	T

LLR	Traces To	Requirement	T/U
LMAP-051	MAP-016	Precursor site markers shall be Node3D nodes in the map scene with area_id metadata.	T
LMAP-052	MAP-012	Resource indicator meshes shall be placed in the map scene at resource locations.	T

4. Player Character

4.1 Scene Structure

LLR	Traces To	Requirement	T/U
LPC-001	PC-001	The player scene root shall be a CharacterBody3D.	T
LPC-002	PC-001	The player scene shall include a CollisionShape3D, Camera3D, HUD (CanvasLayer), and interaction RayCast3D.	T
LPC-003	PC-001	The player scene shall include a scan RayCast3D for journal scanning.	T

4.2 Movement

LLR	Traces To	Requirement	T/U
LPC-010	PC-001	The player shall have movement states: WALK, SPRINT, CROUCH, SWIM, DISABLED.	T
LPC-011	PC-002	SPRINT shall drain stamina at a configurable rate per second.	T
LPC-012	PC-003	When total inventory weight exceeds max_carry, SPRINT shall be disabled.	T
LPC-013	PC-003	When over-encumbered, WALK speed shall be multiplied by a configurable penalty.	T

LLR	Traces To	Requirement	T/U
LPC-014	PC-001	Jump shall apply a vertical velocity impulse.	T
LPC-015	PC-001	CROUCH shall reduce CollisionShape height and reduce speed.	T
LPC-016	PC-001	SWIM shall be triggered by entering a water volume (Area3D).	T

4.3 Weapon Slots

LLR	Traces To	Requirement	T/U
LPC-020	PC-004	The player shall have exactly 2 weapon slots.	T
LPC-021	PC-004	A quick-swap input shall swap the active weapon slot.	T
LPC-022	PC-005	Pistols shall be equippable in the main hand or off-hand.	T
LPC-023	PC-005	Shields shall be equippable only in the off-hand.	T
LPC-024	PC-005	Two-handed weapons, bows, and muskets shall require an empty off-hand.	T

4.4 Death

LLR	Traces To	Requirement	T/U
LPC-030	PC-006	On player_died signal, a DroppedItemBag shall be spawned at the death position.	T
LPC-031	PC-006	The DroppedItemBag shall contain all items from the player's inventory at time of death.	T
LPC-032	PC-008	On spawning a new DroppedItemBag, any previously existing DroppedItemBag shall be destroyed.	T
LPC-033	PC-006	On death, the player shall respawn at the last settlement where they rested.	T

LLR	Traces To	Requirement	T/U
LPC-034	PC-006	On death, the player's inventory shall be cleared.	T
LPC-035	PC-007	On death, the CampInstance shall NOT be destroyed.	T
LPC-036	PC-007	On death, all active hirelings shall NOT be dismissed.	T
LPC-037	PC-009	On death while founding is in progress and not yet reported, founding state shall be cleared.	T
LPC-038	PC-009	On founding cancellation via death, the Surveyor hireling shall return to its origin settlement.	T

4.5 Health

LLR	Traces To	Requirement	T/U
LHP-001	HP-001	PlayerStats shall track health as a float.	T
LHP-002	HP-001	PlayerStats shall track max_health as a float.	T
LHP-003	HP-001	PlayerStats shall emit health_changed when health changes.	T
LHP-004	HP-001	Health shall regenerate at base_health_regen per second (from survival.json).	T
LHP-005	HP-001	Health regen rate shall be multiplied by rest_multiplier when at a settlement or camp.	T
LHP-006	HP-001	When health reaches 0, PlayerStats shall emit player_died.	T

4.6 Stamina

LLR	Traces To	Requirement	T/U
LHP-010	HP-002	PlayerStats shall track stamina as a float.	T
LHP-011	HP-002	PlayerStats shall track max_stamina as a float.	T
LHP-012	HP-002	PlayerStats shall emit stamina_changed when stamina changes.	T
LHP-013	HP-002	Stamina shall regenerate at base_stamina_regen per second when no stamina-consuming action is active.	T
LHP-014	HP-002	Stamina regen rate shall be multiplied by fatigue_modifier from PlayerSurvival.	T
LHP-015	HP-002	Stamina regen rate shall be multiplied by hunger_modifier from PlayerSurvival.	T

4.7 Injury

LLR	Traces To	Requirement	T/U
LINJ-001	HP-003	When the player takes damage while health is below injury_threshold, an injury probability roll shall occur.	T
LINJ-002	HP-003	The injury type shall be selected by weighting against injury probability_by_source for the damage source type.	T
LINJ-003	INJ-001	Deep Wound shall reduce max_health by a configured amount.	T
LINJ-004	INJ-001	Cracked Ribs shall multiply all stamina costs by a configured factor.	T
LINJ-005	INJ-001	Torn Muscle shall multiply attack cooldown by a configured factor.	T

LLR	Traces To	Requirement	T/U
LINJ-006	INJ-001	Bleeding Gash shall apply a health drain DOT at a configured rate.	T
LINJ-007	INJ-001	Venom shall apply a health drain DOT and a vision distortion post-process effect.	T
LINJ-008	INJ-001	Venom shall worsen over time if untreated (DOT rate increases).	T
LINJ-009	INJ-001	Sprained Ankle shall multiply walk and sprint speed by a configured penalty.	T
LINJ-010	INJ-001	Gut Sickness shall reduce stamina regen, add periodic vision wobble, reduce hunger restoration from food, and narrow temperature thresholds.	T
LINJ-011	INJ-002	On sleep, each active injury shall be checked against current rest_tier. If rest_tier >= injury.treatment_tier, the injury shall be removed.	T
LINJ-012	INJ-002	Healer hireling shall add +1 to rest_tier for injury treatment calculations.	T

4.8 Survival

LLR	Traces To	Requirement	T/U
LSURV-001	SURV-001	PlayerSurvival shall track hunger as a float that decreases over time.	T
LSURV-002	SURV-001	PlayerSurvival shall track thirst as a float that decreases over time.	T
LSURV-003	SURV-001	PlayerSurvival shall track temperature as a float.	T
LSURV-004	SURV-001	PlayerSurvival shall track fatigue as a float that increases over time.	T
LSURV-005	SURV-001	When hunger is below warn_threshold, stamina_regen_mult shall be set to 0.5.	T

LLR	Traces To	Requirement	T/U
LSURV-006	SURV-001	When hunger is below crit_threshold, a health DOT shall be applied.	T
LSURV-007	SURV-001	When fatigue exceeds crit_threshold, the player shall be forced to sleep (pass out).	T
LSURV-008	SURV-002	Temperature shall be calculated as: ambient_temp + altitude_mod + time_mod + weather_mod + rain_cold_bonus + clothing_insulation + discipline_mods.	T
LSURV-009	SURV-002	Rain weather shall add a configurable cold bonus to temperature calculation.	T
LSURV-010	SURV-003	Cold weather clothing items shall have weight values defined in items.json.	T
LSURV-011	SURV-003	Cold weather tents shall have a higher weight value than normal tents in items.json.	T
LSURV-012	SURV-004	Interacting with a stream shall give the player an unboiled water item.	T
LSURV-013	SURV-004	Using unboiled water at a campfire shall produce a boiled water item.	T
LSURV-014	SURV-004	Drinking unboiled water shall trigger a Gut Sickness probability roll.	T
LSURV-015	SURV-004	Drinking boiled water shall not trigger a Gut Sickness roll.	T
LSURV-016	SURV-004	Waterskin items shall have weight = 0 when empty.	T
LSURV-017	SURV-004	Waterskin items shall have weight = water_weight when full.	T
LSURV-018	SURV-004	Settlement wells shall provide clean (boiled-equivalent) water.	T
LSURV-019	SURV-001	When total inventory weight exceeds max_carry, the player is over-encumbered.	T

LLR	Traces To	Requirement	T/U
LSURV-020	SURV-001	Encumbrance calculation shall apply Alchemist Light Load modifier to items tagged "plant" or "potion".	T
LSURV-021	SURV-001	Encumbrance calculation shall apply Pathfinder Light Provisions modifier to items tagged "food" or "water".	T

5. Combat

5.1 Melee

LLR	Traces To	Requirement	T/U
LMEL-001	MEL-001	Melee attack shall activate a hitbox Area3D during attack animation frames.	T
LMEL-002	MEL-001	On hitbox body_entered(monster), damage shall be calculated as $\text{weapon.damage} * \text{player_damage_mult} * \text{discipline_mods}$.	T
LMEL-003	MEL-001	Melee attacks shall consume stamina equal to $\text{weapon.stamina_cost}$.	T
LMEL-004	MEL-002	While blocking, incoming damage shall be multiplied by block_reduction .	T
LMEL-005	MEL-003	Shield block_reduction shall be greater than $\text{weapon block_reduction}$.	T
LMEL-006	MEL-002	Each blocked hit shall drain stamina.	T
LMEL-007	MEL-002	Dodge shall trigger an animation with an invulnerability window of N frames.	T
LMEL-008	MEL-002	Dodge shall consume stamina.	T
LMEL-009	MEL-004	Equipping a shield shall prevent equipping a two-handed weapon, bow, or musket.	T

LLR	Traces To	Requirement	T/U
LMEL-010	MEL-003	Shields shall have exactly 1 rune slot.	T

5.2 Bows

LLR	Traces To	Requirement	T/U
LBOW-001	BOW-001	Bows shall have states: IDLE, DRAWING, HELD, RELEASING.	T
LBOW-002	BOW-001	DRAWING shall increase draw_power over time.	T
LBOW-003	BOW-001	HELD shall drain stamina.	T
LBOW-004	BOW-001	RELEASING shall spawn an arrow Rigidbody3D projectile with damage proportional to draw_power.	T
LBOW-005	BOW-001	Aim sway amplitude shall equal $\text{base_sway} * \text{fatigue_mod} * (1.0 - \text{steady_draw_mod})$.	T
LBOW-006	BOW-001	Bows shall have exactly 2 rune slots.	T
LBOW-007	BOW-002	Arrows shall be inventory items with weight.	T
LBOW-008	BOW-002	Firing a bow shall consume 1 arrow from inventory.	T
LBOW-009	BOW-002	Arrows that hit the world shall stick (become StaticBody3D).	T
LBOW-010	BOW-003	Bow firing shall not trigger the monster alert system.	T
LBOW-011	BOW-002	Without Survivalist Arrow Recovery, stuck arrows shall not be recoverable.	T

5.3 Firearms

LLR	Traces To	Requirement	T/U
LRNG-001	RNG-001	Firearms shall have states: READY, FIRING, RELOADING.	T
LRNG-002	RNG-001	FIRING shall use hitscan, consume 1 ammo, and play a loud sound.	T
LRNG-003	RNG-001	RELOADING duration shall equal $\text{base_reload} * (1.0 - \text{quick_load_mod})$.	T
LRNG-004	RNG-001	Muskets shall have exactly 2 rune slots.	T
LRNG-005	RNG-001	Pistols shall have exactly 1 rune slot.	T
LRNG-006	RNG-002	On firearm sound, all monsters within <code>alert_radius</code> shall enter ALERT state.	T
LRNG-007	RNG-003	Ammo crafting recipes shall require mineral powder as an ingredient.	T
LRNG-008	RNG-003	Mineral powder shall be purchasable at settlement shops.	T
LRNG-009	RNG-003	Mineral powder shall be craftable only via Artificer Mechanical Insight ability.	T
LRNG-010	RNG-004	Firearms shall have a condition float (0.0-1.0) that decreases per shot fired.	T
LRNG-011	RNG-004	When condition is below a threshold, each shot shall have a <code>misfire_chance</code> .	T
LRNG-012	RNG-004	Firearm condition shall be restorable at a settlement smithy or via Artificer Field Repair.	T
LRNG-013	RNG-001	Firearm aim sway amplitude shall equal $\text{base_sway} * \text{fatigue_mod} * (1.0 - \text{steady_hands_mod})$.	T

5.4 Runes

LLR	Traces To	Requirement	T/U
LRUNE-001	RUNE-003	On rune generation, effect_value shall be sampled from a Poisson distribution with mean = rune.poisson_mean.	T
LRUNE-002	RUNE-003	Effect_value shall be clamped to [poisson_min, poisson_max].	T
LRUNE-003	RUNE-003	Higher rarity runes shall have higher poisson_mean values.	T
LRUNE-004	RUNE-004	Runes with the life_steal tag shall be rejected when socketed into bows, muskets, or pistols.	T
LRUNE-005	RUNE-003	Each rune instance's rolled effect_value shall be stored in the save file.	T

5.5 Monster AI

LLR	Traces To	Requirement	T/U
LMAI-001	MON-004	MonsterBase shall be a CharacterBody3D with NavigationAgent3D and a StateMachine.	T
LMAI-002	MON-004	MonsterBase StateMachine states shall include: IDLE, PATROL, ALERT, CHASE, ATTACK, FLEE, DESPAWN.	T
LMAI-003	MON-004	Territorial behavior: PATROL within home_radius. On player detection: CHASE. At melee range: ATTACK. Below flee_hp: FLEE.	T
LMAI-004	MON-004	Ambush behavior: IDLE (hidden). On player within trigger_radius: ATTACK burst. Then CHASE or FLEE.	T
LMAI-005	MON-004	Swarm behavior: shared group aggro. One detects, all CHASE. Below group threshold: remaining FLEE.	T
LMAI-006	MON-004	Heavy behavior: IDLE. On first provocation: ALERT. On second: CHARGE.	T

LLR	Traces To	Requirement	T/U
LMAI-007	MON-004	Apex behavior: multi-phase ATTACK with HP thresholds.	T
LMAI-008	MON-004	Apex monsters shall ignore Ward, Decoy, and Banishment Circle effects.	T
LMAI-009	MON-005	Monsters with fire_reaction="avoid" shall add campfire positions to navigation avoidance.	T
LMAI-010	MON-005	Monsters with fire_reaction="attracted" shall pathfind toward campfire positions.	T
LMAI-011	MON-006	Monsters with light_reaction="attracted" shall pathfind toward active light sources.	T
LMAI-012	MON-006	Monsters with light_reaction="repelled" shall add light source positions to navigation avoidance.	T
LMAI-013	MON-001	Monsters beyond despawn_distance from the player shall queue_free.	T

6. Spawn System

LLR	Traces To	Requirement	T/U
LSPN-001	MON-001	A SpawnManager autoload shall trigger spawns when the player enters an area.	T
LSPN-002	MON-001	Spawn count shall be weighted by: weight * time_modifier * difficulty_scalar * (1.0 - suppression_pct).	T
LSPN-003	MON-002	No monsters shall spawn within safe_zone_radius of a settlement_site position.	T
LSPN-004	MON-003	Suppression percentage shall be read from the settlement's tier data.	T
LSPN-005	MON-001	Spawn positions shall use Marker3D nodes in the map scene or NavMesh sampling outside safe zone.	T

7. Navigation and Exploration

7.1 Compass and Map

LLR	Traces To	Requirement	T/U
LNAV-001	NAV-002	A CompassHUD element shall display the cardinal direction the player is facing.	T
LNAV-002	NAV-002	The compass shall be always visible on the HUD.	T
LNAV-003	NAV-001	The player's map shall start with zero revealed hexes.	T
LNAV-004	NAV-001	Hexes shall be revealable by: surveying tools, Cartographer hireling, purchased maps, or Pathfinder Cartography ability.	T
LNAV-005	NAV-009	The MapViewer shall never render the player's position.	T

7.2 Surveying

LLR	Traces To	Requirement	T/U
LNAV-010	NAV-004	Surveying tools shall reveal hexes in a radius when used above a minimum elevation.	T
LNAV-011	NAV-004	Naked eye reveal radius shall be 2 hexes.	T
LNAV-012	NAV-004	Spyglass reveal radius shall be 4 hexes. Purchasable at Trading Post+.	T
LNAV-013	NAV-004	Theodolite reveal radius shall be 6 hexes. Purchasable at Town+.	T
LNAV-014	NAV-004	Pathfinder Wayfinder shall multiply surveying radius by 1.5.	T

7.3 Purchased Maps

LLR	Traces To	Requirement	T/U
LNAV-020	NAV-005	Village shops shall sell maps that reveal all hexes within 2-hex radius of the settlement.	T
LNAV-021	NAV-005	Town shops shall sell maps that reveal 3-hex radius.	T
LNAV-022	NAV-005	City shops shall sell maps that reveal 4-hex radius.	T
LNAV-023	NAV-005	Purchased maps shall show biomes, rivers, cliffs, and landmarks within the revealed area.	T

7.4 Revealed Hex Bonus

LLR	Traces To	Requirement	T/U
LNAV-025	NAV-006	When a hex is revealed, all landmarks within 2 hexes beyond the revealed hex shall be added to a visible_landmarks list.	T
LNAV-026	NAV-006	Visible landmarks from the bonus shall appear on the map without revealing their hex's terrain data.	T

7.5 Map Markers

LLR	Traces To	Requirement	T/U
LNAV-030	NAV-008	The player shall be able to place markers on the MapViewer at hex positions.	T
LNAV-031	NAV-008	Marker types shall include: fox (hunting), campsite, portage, rapids, and custom.	T
LNAV-032	NAV-008	The player shall be able to remove placed markers.	T

LLR	Traces To	Requirement	T/U
LNAV-033	NAV-008	Player markers shall persist in the save file.	T

7.6 Scanning and Journal

LLR	Traces To	Requirement	T/U
LNAV-040	NAV-011	A scan input shall cast the scan RayCast3D.	T
LNAV-041	NAV-011	If the raycast hits a node in the "scannable" group, a journal entry shall be added for that object.	T
LNAV-042	NAV-011	Scannable categories shall include: creatures, landmarks, items, and resources.	T
LNAV-043	NAV-011	Scanning shall provide visual and audio feedback on success.	T
LNAV-044	NAV-013	Landmarks shall be nameable on first scan or first Area3D trigger entry.	T

7.7 Resource Discovery

LLR	Traces To	Requirement	T/U
LNAV-050	NAV-012	Each resource indicator in the map shall have an Area3D proximity trigger.	T
LNAV-051	NAV-012	On player entering a resource trigger, the resource_id shall be added to AreaManager.discovered_resources for that area.	T
LNAV-052	NAV-012	Duplicate discoveries shall be ignored.	T
LNAV-053	NAV-012	Surveyor hireling shall call AreaManager.discover_all(area_id) on sleep.	T

8. Settlement Founding

LLR	Traces To	Requirement	T/U
LFOUND-001	FOUND-001	A FoundingManager shall track founding state.	T
LFOUND-002	FOUND-001	Founding shall require an active Surveyor hireling.	T
LFOUND-003	FOUND-001	No flag items shall exist in the game.	T
LFOUND-004	FOUND-002	On camp placement in an unsettled area with a Surveyor, the game shall display a directional hint toward the settlement site.	U
LFOUND-005	FOUND-002	Camp setup shall play a 2-second cutscene at the settlement site.	T
LFOUND-006	FOUND-003	After directing the Surveyor, a timer of 1-2 in-game days (configurable) shall begin.	T
LFOUND-007	FOUND-003	While the timer is running, the Surveyor shall remain at the camp.	T
LFOUND-008	FOUND-004	After the Surveyor completes, the player must interact with the mayor NPC at the origin Village.	T
LFOUND-009	FOUND-005	On the next sleep after reporting, a SettlementInstance shall be created at the settlement_site with tier=1 and trade_score=0.	T
LFOUND-010	FOUND-005	On founding, the new settlement shall run its initial road evaluation.	T
LFOUND-011	FOUND-006	If the player dies before reporting, all founding state shall be cleared.	T
LFOUND-012	FOUND-006	On founding cancellation, the Surveyor shall return to the origin settlement.	T

9. Camps and Hirelings

9.1 Camps

LLR	Traces To	Requirement	T/U
LCAMP-001	CAMP-001	On camp request, the game shall display a directional hint toward a valid camp spot.	U
LCAMP-002	CAMP-001	The player shall walk to the camp spot (no teleportation).	T
LCAMP-003	CAMP-001	On arrival at the camp spot, a 2-second setup cutscene shall play.	T
LCAMP-004	CAMP-001	After the cutscene, a CampInstance node shall be spawned.	T
LCAMP-005	CAMP-002	CampfireNode shall have an Area3D with a monster_deterrent_radius.	T
LCAMP-006	CAMP-003	Campfire position shall be added to navigation avoidance for monsters with fire_reaction="avoid".	T
LCAMP-007	CAMP-003	Monsters with fire_reaction="attracted" shall pathfind toward campfire position.	T
LCAMP-008	CAMP-002	CampfireNode shall provide a warmth modifier within its radius.	T
LCAMP-009	CAMP-002	CampfireNode shall enable cooking interaction.	T
LCAMP-010	CAMP-002	CampfireNode shall be a light source (OmniLight3D).	T
LCAMP-011	CAMP-005	An "Abandon Camp" menu option shall dismiss all hirelings and destroy the CampInstance.	T
LCAMP-012	CAMP-005	On abandon, all camp items shall be destroyed.	T

9.2 Hirelings

LLR	Traces To	Requirement	T/U
LHIRE-001	HIRE-001	HirelingManager shall track active_hirelings as an Array.	T
LHIRE-002	HIRE-001	Each hireling shall have: type, cost_per_day, origin_settlement.	T
LHIRE-003	HIRE-001	Hirelings shall spawn tent visuals at the camp.	T
LHIRE-004	HIRE-002	Hirelings shall be excluded from monster collision/targeting layers.	T
LHIRE-005	HIRE-002	Hirelings shall play a cower animation when a monster enters ATTACK state near camp.	T
LHIRE-006	HIRE-002	Hirelings shall not be killable.	T
LHIRE-007	HIRE-003	Porter shall spawn a PorterBackpack container at camp with capacity from hirelings.json.	T
LHIRE-008	HIRE-003	PorterBackpack contents shall persist when camp moves.	T
LHIRE-009	HIRE-003	When at a player home, Porter shall enable a transfer UI between backpack and home storage.	T
LHIRE-010	HIRE-004	Hunter shall generate food items on sleep. Quantity from hirelings.json.	T
LHIRE-011	HIRE-005	Guide shall display directional indicators toward POIs within guide_radius.	T
LHIRE-012	HIRE-006	Surveyor shall call discover_all(current_area_id) on sleep.	T
LHIRE-013	HIRE-006	Surveyor shall be required for founding (gate check in FoundingManager).	T

LLR	Traces To	Requirement	T/U
LHIRE-014	HIRE-007	Cartographer shall track area_ids entered while active.	T
LHIRE-015	HIRE-007	On rest, Cartographer shall generate a MapItem covering tracked areas.	T
LHIRE-016	HIRE-008	Healer shall set camp rest_tier to 3 (Town equivalent) for injury treatment.	T
LHIRE-017	HIRE-009	On sleep, cost_per_day shall be deducted per active hireling.	T
LHIRE-018	HIRE-009	If funds are insufficient for a hireling's cost, that hireling shall depart.	T

10. Economy and Settlements

10.1 Runtime

LLR	Traces To	Requirement	T/U
LECO-001	TIER-001	A SettlementManager autoloader shall maintain a Dict of SettlementState per settlement.	T
LECO-002	TIER-001	SettlementState shall include: area_id, tier, trade_score, is_port, connections[], road_budget_spent, visual_instance.	T
LECO-003	TIER-006	On tier change, the settlement visual scene shall be swapped to match the new tier.	T
LECO-004	TIER-001	NPC scenes shall be instantiated per tier services list.	T
LECO-005	TIER-003	When a settlement on a RiverArea reaches tier 3 (Town), adjacent river edges shall be marked passable (bridge).	T
LECO-006	TIER-002	Trading Post shall exploit resources from own Area only (100%).	T

LLR	Traces To	Requirement	T/U
LECO-007	TIER-002	Village shall exploit own Area (100%) + adjacent Areas (50%).	T
LECO-008	TIER-002	Town shall exploit own Area (100%) + adjacent Areas (100%).	T
LECO-009	TIER-002	City shall exploit own (100%) + adjacent (100%) + ring-2 (25%).	T
LECO-010	TIER-005	Crestport shall start as a Village (tier 2) in a Deep Water Port Area.	T

10.2 Trade Routing

LLR	Traces To	Requirement	T/U
LTRADE-001	PORT-002	A TradeGraph shall maintain an adjacency list of settlements connected by roads.	T
LTRADE-002	PORT-002	On sleep, Dijkstra shall be run from each non-port settlement to find the nearest port.	T
LTRADE-003	PORT-002	The route_path for each settlement shall be stored.	T
LTRADE-004	PORT-003	On new settlement founded or new road formed, trade_graph_dirty shall be set true.	T
LTRADE-005	PORT-003	On sleep, if dirty, all routes shall be recalculated.	T

10.3 Consumption and Trade Score

LLR	Traces To	Requirement	T/U
LSCORE-001	GROW-001	Local output for a settlement shall be the sum of (resource.richness * exploitation_pct) for own + adjacent + ring-2 per tier.	T

LLR	Traces To	Requirement	T/U
LSCORE-002	GROW-003	For each source settlement, trace route to port. At each node, remaining_value shall be multiplied by (1.0 - node.consumption_rate).	T
LSCORE-003	GROW-003	Each node's consumed portion (the amount removed by consumption) shall contribute to that node's trade_score.	T
LSCORE-004	GROW-004	Consumption shall apply only to raw materials flowing coastward.	T
LSCORE-005	GROW-004	Manufactured goods availability shall NOT be reduced by consumption.	T
LSCORE-006	GROW-001	trade_score shall accumulate per sleep: += (local_output * local_weight) + (consumed_passthrough * passthrough_weight) + (diversity_curve[count]) + (connections * connection_bonus).	T
LSCORE-007	GROW-002	Tier advancement shall require: trade_score >= next_tier.score_threshold.	T
LSCORE-008	GROW-002	Tier advancement shall require: all required_resources present in flow-through set.	T
LSCORE-009	GROW-002	On tier advancement, emit tier_changed signal.	T
LSCORE-010	GROW-002	On tier_changed, trigger road re-evaluation for that settlement.	T

10.4 Export

LLR	Traces To	Requirement	T/U
LEXPORT-001	PORT-004	Global export value shall be the sum of remaining_value at ports across all trade routes.	T
LEXPORT-002	PORT-004	The highest exceeded threshold in export_thresholds.json shall determine available_goods_tier.	T
LEXPORT-003	PORT-005	Shop inventories shall filter goods by: (a) export_tier <= global tier, (b) min_settlement_tier <= this settlement tier.	T

10.5 Roads

LLR	Traces To	Requirement	T/U
LROAD-001	ROAD-001	Every settlement shall have a road_budget of 9.	T
LROAD-002	ROAD-002	Road cost shall equal the hex distance between the two settlements.	T
LROAD-003	ROAD-003	Road evaluation shall occur on settlement founding.	T
LROAD-004	ROAD-003	Road evaluation shall occur on settlement tier change.	T
LROAD-005	ROAD-003	Road evaluation shall NOT occur on every sleep.	T
LROAD-006	ROAD-003	During evaluation, settlements shall be sorted by trade_score descending.	T
LROAD-007	ROAD-003	The settlement shall connect to the highest-scoring reachable target it can afford, then the next, until budget is exhausted.	T
LROAD-008	ROAD-004	Only the initiating settlement shall spend road budget. The target shall receive the connection at no cost.	T
LROAD-009	ROAD-005	Old roads shall never be deleted on re-evaluation.	T
LROAD-010	ROAD-005	Available budget on re-evaluation shall be 9 minus the total cost of existing outgoing roads.	T
LROAD-011	ROAD-007	Road pathfinding shall use A* on the hex grid.	T
LROAD-012	ROAD-007	Road paths shall not cross cliff edges.	T
LROAD-013	ROAD-007	Road paths shall not cross river edges unless a Ford exists or the settlement on the river Area is Town+.	T

LLR	Traces To	Requirement	T/U
LROAD-014	ROAD-007	Road paths shall not pass through Mountain-tagged areas unless they have Mountain Pass.	T
LROAD-015	ROAD-008	Maximum road length shall be 9 hexes.	T
LROAD-016	ROAD-006	Road quality shall equal the minimum tier of the two connected settlements.	T
LROAD-017	ROAD-006	Road quality shall auto-upgrade when either endpoint tiers up.	T
LROAD-018	ROAD-006	RoadVisual mesh/material shall be selected from roads.json based on quality.	T
LROAD-019	ROAD-006	A speed zone Area3D along road paths shall apply a speed multiplier to the player.	T

10.6 Shops and Homes

LLR	Traces To	Requirement	T/U
LSHOP-001	PORT-005	PlayerStats shall track gold as an integer.	T
LSHOP-002	PORT-005	Selling items at shop NPCs shall increase gold.	T
LSHOP-003	PORT-005	Purchasing items shall decrease gold.	T
LHOME-001	HOME-001	Homes shall be purchasable at Village+ settlements.	T
LHOME-002	HOME-001	Home storage shall be a single global store shared across all owned homes.	T
LHOME-003	HOME-001	Home storage capacity shall be unlimited.	T

LLR	Traces To	Requirement	T/U
LHOME-004	HOME-002	When a Porter hireling is at a player's home, a transfer UI between Porter backpack and home storage shall be available.	T
LHOME-005	HOME-001	Homes shall include an alchemy workshop interaction.	T
LHOME-006	HOME-001	Homes shall provide a free bed (no cost to sleep).	T

11. Sleep and Batch

LLR	Traces To	Requirement	T/U
LSLEEP-001	SLEEP-001	Sleep shall be triggerable at a settlement bed or camp tent.	T
LSLEEP-002	SLEEP-001	Sleep shall advance the clock to the next morning.	T
LSLEEP-003	SLEEP-002	Batch step 0: process founding completion (if founding_pending, create settlement).	T
LSLEEP-004	SLEEP-002	Batch step 1: process resource extraction (accumulate richness values per settlement per exploitation tier).	T
LSLEEP-005	SLEEP-002	Batch step 2: if trade_graph_dirty, recalculate all trade routes via Dijkstra.	T
LSLEEP-006	SLEEP-002	Batch step 3: calculate consumption along routes and update per-settlement trade scores.	T
LSLEEP-007	SLEEP-002	Batch step 4: check tier advancement for all settlements.	T
LSLEEP-008	SLEEP-002	Batch step 5: update settlement visuals for any tier changes.	T
LSLEEP-009	SLEEP-002	Batch step 6: run road evaluation for any settlements that tiered up.	T

LLR	Traces To	Requirement	T/U
LSLEEP-010	SLEEP-002	Batch step 7: recalculate Area suppression percentages.	T
LSLEEP-011	SLEEP-002	Batch step 8: generate Hunter food if Hunter hireling active.	T
LSLEEP-012	SLEEP-002	Batch step 9: process Trapper harvests if Trapper traps placed.	T
LSLEEP-013	SLEEP-002	Batch step 10: deduct hireling daily costs. Dismiss if insufficient funds.	T
LSLEEP-014	SLEEP-002	Batch step 11: process injury healing based on rest_tier.	T
LSLEEP-015	SLEEP-002	Batch step 12: reset fatigue to 0.	T
LSLEEP-016	SLEEP-002	Batch step 13: deduct hunger and thirst for elapsed time.	T
LSLEEP-017	SLEEP-002	Batch step 14: advance day counter. Check season advancement.	T
LSLEEP-018	SLEEP-002	Batch step 15: roll new weather.	T
LSLEEP-019	SLEEP-002	Batch step 16: process Surveyor discovery if active.	T
LSLEEP-020	SLEEP-002	Batch step 17: process Cartographer map generation if active.	T
LSLEEP-021	SLEEP-002	Batch step 18: auto-save.	T
LSLEEP-022	SLEEP-003	An optional SleepSummaryUI shall display events from the batch (tier changes, roads, hirelings departed, injuries healed).	T

12. Disciplines

12.1 System

LLR	Traces To	Requirement	T/U
LDSYS-001	DSYS-002	DisciplineManager shall track attunements as an Array with max size 3.	T
LDSYS-002	DSYS-002	Attunements shall be permanent and non-respecable.	T
LDSYS-003	DSYS-004	Interacting with a Place of Power when attunements < 3 shall show an attunement UI.	T
LDSYS-004	DSYS-004	On attunement confirm, the discipline shall be added and ability 1 unlocked.	T
LDSYS-005	DSYS-004	Interacting with a Place of Power when attunements == 3 shall add a journal entry only.	T
LDSYS-006	DSYS-005	DisciplineManager shall subscribe to SignalBus events matching xp_triggers from disciplines.json.	T
LDSYS-007	DSYS-005	On matching signal, the corresponding discipline's XP shall increase.	T
LDSYS-008	DSYS-006	A City Enhancement NPC shall display: available disciplines, current XP, next ability cost.	T
LDSYS-009	DSYS-006	On enhancement purchase, XP and gold shall be deducted and unlocked_abilities incremented.	T
LDSYS-010	DSYS-007	Visiting a second Place of Power of the same discipline shall grant bonus XP.	T

12.2 Survivalist Abilities

LLR	Traces To	Requirement	T/U
LSURV-A01	DSYS-008	Trapper: placing a TrapNode in the world shall be possible via the ability.	T

LLR	Traces To	Requirement	T/U
LSURV-A02	DSYS-008	Trapper: on sleep, each placed trap shall roll food_chance and add food to inventory on success.	T
LSURV-A03	DSYS-008	Arrow Recovery: on combat_ended signal, floor(arrows_fired * recovery_pct) arrows shall be added to inventory.	T
LSURV-A04	DSYS-008	Night Eyes: toggling shall enable a greyscale post-process shader.	T
LSURV-A05	DSYS-008	Night Eyes: while active, stamina shall drain at night_eyes_drain_rate/sec.	T
LSURV-A06	DSYS-008	Light Foot: player stealth_modifier shall be set, reducing monster detection_range.	T
LSURV-A07	DSYS-008	Light Foot: bow attacks on unaware monsters shall deal damage * stealth_mult.	T
LSURV-A08	DSYS-008	Steady Draw: bow draw_speed shall be multiplied by steady_draw_speed_mult.	T
LSURV-A09	DSYS-008	Steady Draw: bow aim sway shall be reduced by steady_draw_sway_reduction.	T
LSURV-A10	DSYS-008	Steady Draw: held draw stamina drain shall be multiplied by steady_draw_drain_mult.	T
LSURV-A11	DSYS-008	Weatherskin: cold_threshold and heat_threshold shall be widened by weatherskin_range.	T
LSURV-A12	DSYS-008	Endurance: max_stamina shall be multiplied by 1.25.	T
LSURV-A13	DSYS-008	Endurance: fatigue stamina penalty shall be multiplied by 0.5.	T
LSURV-A14	DSYS-008	Ghost Walk: on activation, player shall be invisible to monsters for 10 seconds.	T
LSURV-A15	DSYS-008	Ghost Walk: shall have a long cooldown timer.	T

LLR	Traces To	Requirement	T/U
LSURV-A16	DSYS-008	Ghost Walk: shall break on attack or interaction.	T

12.3 Ritualist Abilities

LLR	Traces To	Requirement	T/U
LRIT-A01	DSYS-008	Ward: 5-second channel shall be interruptible by damage.	T
LRIT-A02	DSYS-008	Ward: on completion, spawn a WardZone Area3D at target with 10-foot radius.	T
LRIT-A03	DSYS-008	Ward: monsters shall not enter the WardZone (added to navigation avoidance).	T
LRIT-A04	DSYS-008	Ward: WardZone shall last 60 seconds.	T
LRIT-A05	DSYS-008	Ward: cooldown shall be 15 seconds.	T
LRIT-A06	DSYS-008	Preserve: all food spoil timers shall be multiplied by preserve_decay_mult (>1.0, slowing decay).	T
LRIT-A07	DSYS-008	Preserve: weapon and armor durability loss per use shall be multiplied by preserve_durability_mult (<1.0).	T
LRIT-A08	DSYS-008	Decoy: 5-second channel. On completion, spawn a DecoyCircle at target.	T
LRIT-A09	DSYS-008	Decoy: after 10-second delay, spawn a visible decoy mesh and AudioStreamPlayer3D that generates noise.	T
LRIT-A10	DSYS-008	Decoy: non-Apex monsters within detection range shall pathfind toward the decoy.	T
LRIT-A11	DSYS-008	Decoy: shall persist for 30 seconds before fading.	T

LLR	Traces To	Requirement	T/U
LRIT-A12	DSYS-008	Enchant: 5-second channel. On completion, apply enchant_damage_bonus to equipped weapon for 60 seconds.	T
LRIT-A13	DSYS-008	Enchant: shall add a visible glow (OmniLight3D) to the weapon.	T
LRIT-A14	DSYS-008	Enchant: bonus shall stack with existing rune effects.	T
LRIT-A15	DSYS-008	Sending Chest: 5-second channel. On completion, spawn an interactable ChestNode.	T
LRIT-A16	DSYS-008	Sending Chest: on chest close, all contents shall transfer to HomeStorage.	T
LRIT-A17	DSYS-008	Sending Chest: the chest shall queue_free after transfer.	T
LRIT-A18	DSYS-008	Sending Chest: shall be usable once per rest.	T
LRIT-A19	DSYS-008	Banishment Circle: 5-second channel. On completion, spawn a BanishZone Area3D trap.	T
LRIT-A20	DSYS-008	Banishment Circle: first non-Apex monster to enter BanishZone shall be despawned.	T
LRIT-A21	DSYS-008	Banishment Circle: BanishZone shall queue_free after triggering.	T
LRIT-A22	DSYS-008	Banishment Circle: BanishZone shall persist until triggered or next rest.	T
LRIT-A23	DSYS-008	Anchor part 1: 5-second channel. Store anchor_position in save.	T
LRIT-A24	DSYS-008	Anchor part 2: 5-second channel from any location. Teleport player to anchor_position.	T
LRIT-A25	DSYS-008	Anchor: on teleport, anchor_position shall be cleared (one use).	T

12.4 Pathfinder Abilities

LLR	Traces To	Requirement	T/U
LPATH-A01	DSYS-009	Wayfinder: surveying_range_mult shall be set to 1.5.	T
LPATH-A02	DSYS-009	Light Provisions: items tagged "food" or "water" shall have weight multiplied by 0.5.	T
LPATH-A03	DSYS-009	Cartography: at camp, the player shall be able to generate a MapItem from areas entered since last rest.	T
LPATH-A04	DSYS-009	Prospect: on activation, display a directional indicator toward the nearest undiscovered resource in the current Area.	T
LPATH-A05	DSYS-009	Steady Pace: stamina_regen_mult shall be multiplied by 1.3.	T
LPATH-A06	DSYS-009	Pathsense: precursor sites within pathsense_radius shall render a shimmer VFX when viewed from above elevation threshold.	U

12.5 Warrior Abilities

LLR	Traces To	Requirement	T/U
LWAR-A01	DSYS-008	Heavy Hand: melee_damage_mult shall be multiplied by 1.2.	T
LWAR-A02	DSYS-008	Pack Mule: max_carry shall be multiplied by 1.3.	T
LWAR-A03	DSYS-008	Pack Mule: over-encumbrance speed penalty shall be reduced.	T
LWAR-A04	DSYS-008	Ironclad: can_equip_heavy flag shall be set true.	T
LWAR-A05	DSYS-008	Ironclad: all armor damage_reduction shall be increased by ironclad_bonus.	T
LWAR-A06	DSYS-008	Stagger: charged attack shall set non-Apex targets to STAGGERED state.	T

LLR	Traces To	Requirement	T/U
LWAR-A07	DSYS-008	Stagger: shall consume high stamina.	T
LWAR-A08	DSYS-008	Second Wind: shall only activate when health < 25%.	T
LWAR-A09	DSYS-008	Second Wind: shall restore 30% stamina and grant 8s damage resistance.	T
LWAR-A10	DSYS-008	Second Wind: shall be usable once per rest.	T
LWAR-A11	DSYS-008	Warcry: all non-Apex monsters within 15m shall enter FLINCH state.	T
LWAR-A12	DSYS-008	Warcry: shall have a moderate cooldown.	T

12.6 Swordsman Abilities

LLR	Traces To	Requirement	T/U
LSWD-A01	DSYS-008	Riposte: dodging during a monster's attack window shall enable a riposte window.	T
LSWD-A02	DSYS-008	Riposte: next attack during the window shall deal 2x damage.	T
LSWD-A03	DSYS-008	Cripple: riposte hits shall apply a cripple debuff (speed reduction or attack disable) lasting 15 seconds.	T
LSWD-A04	DSYS-008	Read: monster attack telegraph animations shall play slower (configured multiplier).	T
LSWD-A05	DSYS-008	Read: dodge invulnerability window shall be extended by configured frames.	T
LSWD-A06	DSYS-008	Off-Hand Mastery (shield): block_reduction shall receive a bonus.	T
LSWD-A07	DSYS-008	Off-Hand Mastery (shield): block recovery time shall be reduced.	T

LLR	Traces To	Requirement	T/U
LSWD-A08	DSYS-008	Off-Hand Mastery (pistol): weapon swap time shall be reduced.	T
LSWD-A09	DSYS-008	Off-Hand Mastery (pistol): aim bonus shall be applied after a melee combo.	T
LSWD-A10	DSYS-008	Bleed: critical hits shall apply a BleedDOT dealing damage over 10 seconds.	T
LSWD-A11	DSYS-008	Bleed: shall stack up to 3 times.	T
LSWD-A12	DSYS-008	Bleed: Swordsman bleeds shall have increased duration and damage vs baseline.	T
LSWD-A13	DSYS-008	Blade Sense: Bleeding Gash injury probability shall be multiplied by (1.0 - blade_sense_reduction).	T
LSWD-A14	DSYS-008	Blade Sense: Deep Wound injury probability shall be multiplied by (1.0 - blade_sense_reduction).	T
LSWD-A15	DSYS-008	Flurry: shall only activate during a riposte window.	T
LSWD-A16	DSYS-008	Flurry: shall trigger a 3-strike chain animation.	T
LSWD-A17	DSYS-008	Flurry: shall consume heavy stamina.	T
LSWD-A18	DSYS-008	Deathmark: cripple_count per monster shall be tracked per encounter.	T
LSWD-A19	DSYS-008	Deathmark: at 3+ cripples, monster.damage_taken_mult shall be set to 1.3.	T

12.7 Artificer Abilities

LLR	Traces To	Requirement	T/U
LART-	DSYS-	Field Repair: at camp, the player shall select a damaged item and consume	T

LLR	Traces To	Requirement	T/U
A01	008	materials to restore durability.	
LART-A02	DSYS-008	Quick Load: reload_time_mult shall be multiplied by 0.7.	T
LART-A03	DSYS-008	Ammo Smith: at camp, craft ammo from iron + mineral powder per recipe.	T
LART-A04	DSYS-008	Spike Trap: craft from iron + wood. Place TrapNode in world.	T
LART-A05	DSYS-008	Spike Trap: on monster enter, deal spike_damage.	T
LART-A06	DSYS-008	Spike Trap: if not triggered, the trap shall be recoverable via interaction.	T
LART-A07	DSYS-008	Steady Hands: aim_sway_fatigue_mult shall be multiplied by 0.5.	T
LART-A08	DSYS-008	Reinforce: at camp, select weapon or armor. Apply temp buff (damage or armor bonus) lasting until next sleep.	T
LART-A09	DSYS-008	Reinforce: shall consume materials.	T
LART-A10	DSYS-008	Jury Rig: in the field (no camp required), select a broken item. Restore to temporary durability (~10 uses).	T
LART-A11	DSYS-008	Jury Rig: shall not require materials.	T
LART-A12	DSYS-008	Mechanical Insight: player.can_activate_precursor shall be set true.	T
LART-A13	DSYS-008	Mechanical Insight: precursor mechanism nodes shall check this flag before allowing interaction.	T
LART-A14	DSYS-008	Mechanical Insight: the player shall be able to disassemble precursor artifacts into rare materials.	T
LART-A15	DSYS-008	Mechanical Insight: the player shall be able to craft mineral powder.	T

12.8 Alchemist Abilities

LLR	Traces To	Requirement	T/U
LALC-A01	DSYS-008	Keen Eye: hidden GatherNodes shall have their visibility set to true.	T
LALC-A02	DSYS-008	Light Load: items tagged "plant" or "potion" shall have weight multiplied by 0.5.	T
LALC-A03	DSYS-008	Extended Potency: active potion buff durations shall be multiplied by 1.5.	T
LALC-A04	DSYS-008	Toxicologist: poison damage received shall be multiplied by 0.5.	T
LALC-A05	DSYS-008	Toxicologist: Venom injury treatment_tier shall be reduced by 1.	T
LALC-A06	DSYS-008	Master Brewer: recipes flagged requires_master_brewer in recipes.json shall become available.	T
LALC-A07	DSYS-008	Master Brewer: firefly jar shall be included in unlockable recipes.	T
LALC-A08	DSYS-008	Identify: interacting with an unknown material shall reveal its properties in the journal.	T
LALC-A09	DSYS-008	Identify: examining a material shall unlock associated experimental recipes.	T
LALC-A10	DSYS-008	Alchemist shall be able to purify unboiled water (removes Gut Sickness risk without campfire boiling).	T

12.9 Scholar Abilities

LLR	Traces To	Requirement	T/U
LSCH-A01	DSYS-010	Runic Literacy: precursor inscription nodes shall display translated text for Scholar players.	T

LLR	Traces To	Requirement	T/U
LSCH-A02	DSYS-010	Runic Literacy: non-Scholar players shall see gibberish texture/mesh for the same nodes.	T
LSCH-A03	DSYS-010	Salvage: loot quantities from precursor sites shall be multiplied by salvage_mult.	T
LSCH-A04	DSYS-010	Shelter: sleeping at a precursor ruin site shall set rest_tier to 2 (Village equivalent).	T
LSCH-A05	DSYS-010	Shelter: sleeping at a ruin shall provide weather protection.	T
LSCH-A06	DSYS-010	Glyph Ward: at precursor sites with glyph nodes, the player shall be able to activate them.	T
LSCH-A07	DSYS-010	Glyph Ward: activated glyphs shall create a safe zone Area3D with larger radius than Ritualist Ward.	T
LSCH-A08	DSYS-010	Glyph Ward: safe zone shall last longer than Ritualist Ward.	T
LSCH-A09	DSYS-010	Glyph Ward: shall only function at precursor sites (not placeable anywhere).	T
LSCH-A10	DSYS-010	Deep Reading: interacting with a precursor artifact shall reveal related precursor site hex locations on the player's map.	T
LSCH-A11	DSYS-010	Reactivate: at larger ruins with sustenance_system nodes, the player shall be able to activate the system.	T
LSCH-A12	DSYS-010	Reactivate: using an active sustenance system shall fully restore hunger and thirst.	T
LSCH-A13	DSYS-010	Reactivate: each system shall have a configurable max_charges.	T
LSCH-A14	DSYS-010	Reactivate: each use shall decrement charges by 1.	T
LSCH-A15	DSYS-010	Reactivate: at 0 charges, the system shall be permanently depleted.	T

LLR	Traces To	Requirement	T/U
LSCH-A16	DSYS-010	Reactivate: system state (active, charges remaining) shall persist in save.	T

12.10 Wizard Abilities

LLR	Traces To	Requirement	T/U
LWIZ-A01	DSYS-008	Each Wizard spell: on use, check for required reagent in inventory. If absent, display fail message.	T
LWIZ-A02	DSYS-008	Each Wizard spell: on use, consume 1 of the required reagent from inventory.	T
LWIZ-A03	DSYS-008	Firebolt: 1-second cast. Spawn fire projectile. On hit: damage + BurnDOT.	T
LWIZ-A04	DSYS-008	Firebolt: shall consume 1 fire herb.	T
LWIZ-A05	DSYS-008	Frost Snap: 1-second cast. Raycast target. Apply ROOT state for 5 seconds. No damage.	T
LWIZ-A06	DSYS-008	Frost Snap: shall consume 1 crystal shard.	T
LWIZ-A07	DSYS-008	Force Push: instant cast. Cone area. Apply knockback and STAGGER to all monsters in cone.	T
LWIZ-A08	DSYS-008	Force Push: shall consume 1 mineral powder.	T
LWIZ-A09	DSYS-008	Lightning Arc: 1.5-second cast. Primary target + chain to up to 2 additional targets within chain_radius.	T
LWIZ-A10	DSYS-008	Lightning Arc: shall consume 1 charged filament.	T
LWIZ-A11	DSYS-008	Stone Shield: 1-second cast. Spawn destructible barrier (StaticBody3D with HP).	T

LLR	Traces To	Requirement	T/U
LWIZ-A12	DSYS-008	Stone Shield: barrier shall absorb incoming damage until HP depleted or 15 seconds elapsed.	T
LWIZ-A13	DSYS-008	Stone Shield: shall consume 2 mineral powder.	T
LWIZ-A14	DSYS-008	Ruin: 2-second cast. Raycast target. Apply massive single-target damage.	T
LWIZ-A15	DSYS-008	Ruin: shall consume 1 precursor core fragment.	T

13. Alchemy and Cooking

LLR	Traces To	Requirement	T/U
LCRAFT-001	CRAFT-001	GatherNode: world Node3D with Area3D trigger. On player interact: add item to inventory.	T
LCRAFT-002	CRAFT-001	Some GatherNodes shall have hidden=true, visible only when Alchemist Keen Eye is active.	T
LCRAFT-003	CRAFT-002	Cooking at campfire: RecipeUI filtered by station_type="campfire".	T
LCRAFT-004	CRAFT-002	Eating raw (uncooked) food shall trigger a Gut Sickness probability roll.	T
LCRAFT-005	CRAFT-003	Alchemy at camp set or home workshop: RecipeUI station_type="alchemy".	T
LCRAFT-006	CRAFT-003	Master Brewer recipes shall only appear if the ability is unlocked.	T
LCRAFT-007	CRAFT-004	Firefly jar: on equip/use, spawn OmniLight3D with white color and large radius.	T
LCRAFT-008	CRAFT-004	Firefly jar light shall be registered as a light source for monster light_reaction checks.	T

LLR	Traces To	Requirement	T/U
LCRAFT-009	CRAFT-005	Food items with a spoil_timer shall tick down over time.	T
LCRAFT-010	CRAFT-005	Expired food: eating shall trigger a Gut Sickness probability roll.	T
LCRAFT-011	CRAFT-005	Ritualist Preserve shall slow spoil_timer decay.	T
LCRAFT-012	CRAFT-003	Potion buffs: on consume, add BuffInstance {effect_id, duration, magnitude} to PlayerBuffs.	T
LCRAFT-013	CRAFT-003	Potion buff duration shall be multiplied by Extended Potency modifier if active.	T

14. Inventory

LLR	Traces To	Requirement	T/U
LINV-001	INV-001	Inventory shall store items as Array[{item_id, quantity, durability, rune_ids[], spoil_timer, state{}}].	T
LINV-002	INV-001	Total weight shall be recalculated on any inventory change.	T
LINV-003	INV-001	Weight calculation shall apply per-item discipline modifiers (Light Load, Light Provisions, waterskin empty weight).	T
LINV-004	INV-004	Rune socketing UI: player shall select equipment, view slots, and socket a rune from inventory.	T
LINV-005	INV-004	On socketing, rune effects shall be applied to the equipment.	T
LINV-006	INV-004	Socketing a life_steal rune into a ranged weapon shall be rejected.	T
LINV-007	INV-002	PorterBackpack: separate container at camp with capacity from hirelings.json.	T

LLR	Traces To	Requirement	T/U
LINV-008	INV-003	HomeStorage: single global Dict<item_id, quantity> accessible at any owned home.	T
LINV-009	INV-003	HomeStorage: no capacity limit.	T
LINV-010	INV-003	Sending Chest deposits shall go to HomeStorage.	T

15. UI

LLR	Traces To	Requirement	T/U
LUI-001	UI-001	HUD shall include: CompassBar, HealthBar, StaminaBar, EquippedWeaponIcon, QuickSwapIndicator, StatusEffectIcons.	T
LUI-002	UI-001	Survival warnings shall use audio cues and screen vignette effects (not HUD bars).	T
LUI-003	UI-002	InventoryScreen shall pause the game (get_tree().paused).	T
LUI-004	UI-002	InventoryScreen shall show: item grid, equipment slots, encumbrance bar, rune socket button.	T
LUI-005	UI-003	MapView shall render revealed hexes with biome coloring.	T
LUI-006	UI-003	MapView shall display settlement icons at settlement positions.	T
LUI-007	UI-003	MapView shall display player-placed markers.	T
LUI-008	UI-003	MapView shall display named landmarks.	T
LUI-009	UI-005	JournalScreen shall have tabs: Creatures, Ingredients, Landmarks, Sites, Notes.	T

LLR	Traces To	Requirement	T/U
LUI-010	UI-005	Notes tab shall support text input (TextEdit).	T
LUI-011	UI-006	SettlementScreen shall show: tier, resources, trade score, connections, growth progress, consumption rate.	T
LUI-012	UI-007	DisciplineScreen shall show: 3 attunement slots, ability icons per discipline, XP progress bar, next unlock cost.	T
LUI-013	UI-007	Locked abilities shall appear grayed out.	T
LUI-014	UI-009	All UI screens shall be navigable via gamepad D-pad/stick with focus system.	T
LUI-015	UI-009	Settings menu shall include: resolution, window mode, graphics quality, keybindings editor, audio sliders (master/music/sfx/ambient), gameplay toggles.	T

16. Audio

LLR	Traces To	Requirement	T/U
LAUD-001	AUD-001	An AudioManager autoload shall manage AudioBus layout: Master, Music, SFX, Ambient.	T
LAUD-002	AUD-001	Biome ambient tracks shall crossfade based on current area biome type.	T
LAUD-003	AUD-002	Each monster type shall have distinct audio for: idle, alert, attack, death.	U
LAUD-004	AUD-002	Monster audio shall use AudioStreamPlayer3D for 3D positional sound.	T
LAUD-005	AUD-003	Bow firing audio shall use a short-range AudioStreamPlayer3D.	T

LLR	Traces To	Requirement	T/U
LAUD-006	AUD-003	Firearm firing audio shall use a large-range AudioStreamPlayer3D.	T
LAUD-007	AUD-003	Firearm audio shall trigger the monster alert system. Bow audio shall not.	T
LAUD-008	AUD-004	Survival threshold audio cues shall play as one-shot sounds on signal.	T
LAUD-009	AUD-005	Settlement ambient audio complexity shall increase with tier.	U
LAUD-010	AUD-005	Event-driven music shall play on: landmark_discovered, precursor_entered, settlement_grew, first_visit_biome.	U

17. Save System

LLR	Traces To	Requirement	T/U
LSAVE-001	SAVE-001	SaveManager.save() shall collect state from all managers into a single Dictionary.	T
LSAVE-002	SAVE-002	Save state shall include: player position and rotation.	T
LSAVE-003	SAVE-002	Save state shall include: player health, stamina, and active injuries.	T
LSAVE-004	SAVE-002	Save state shall include: player survival values (hunger, thirst, temperature, fatigue).	T
LSAVE-005	SAVE-002	Save state shall include: player inventory (items, quantities, durability, rune instances with rolled values, spoil timers, states).	T
LSAVE-006	SAVE-002	Save state shall include: player equipment and socketed runes.	T
LSAVE-007	SAVE-002	Save state shall include: player attunements, XP per discipline, and unlocked abilities.	T

LLR	Traces To	Requirement	T/U
LSAVE-008	SAVE-002	Save state shall include: player journal entries.	T
LSAVE-009	SAVE-002	Save state shall include: player currency (gold).	T
LSAVE-010	SAVE-002	Save state shall include: all settlement states (area_id, tier, trade_score, is_port, connections, road_budget_spent).	T
LSAVE-011	SAVE-002	Save state shall include: all road data.	T
LSAVE-012	SAVE-002	Save state shall include: trade routing graph.	T
LSAVE-013	SAVE-002	Save state shall include: global export value.	T
LSAVE-014	SAVE-002	Save state shall include: per-area discovered resources and suppression.	T
LSAVE-015	SAVE-002	Save state shall include: current season, day, time of day, and active weather.	T
LSAVE-016	SAVE-002	Save state shall include: precursor seed and site assignments.	T
LSAVE-017	SAVE-002	Save state shall include: landmark names.	T
LSAVE-018	SAVE-002	Save state shall include: active hirelings and their state.	T
LSAVE-019	SAVE-002	Save state shall include: camp position and contents (if active).	T
LSAVE-020	SAVE-002	Save state shall include: owned homes.	T
LSAVE-021	SAVE-002	Save state shall include: home storage contents.	T

LLR	Traces To	Requirement	T/U
LSAVE-022	SAVE-002	Save state shall include: dropped item bag position and contents (if exists).	T
LSAVE-023	SAVE-002	Save state shall include: map reveal state (revealed hexes).	T
LSAVE-024	SAVE-002	Save state shall include: player map markers.	T
LSAVE-025	SAVE-002	Save state shall include: founding state (if in progress).	T
LSAVE-026	SAVE-002	Save state shall include: reactivated ruin states (Scholar Reactivate charges).	T
LSAVE-027	SAVE-002	Save state shall include: Ritualist anchor position (if set).	T
LSAVE-028	SAVE-003	Save format shall be JSON.	T

18. Project Structure

LLR	Traces To	Requirement	T/U
LPROJ-001	CORE-001	Scene directory: res://scenes/ with subdirs: player/, monsters/, settlements/, camps/, ui/, map/, precursor/.	T
LPROJ-002	CORE-001	Script directory: res://scripts/ with subdirs: autoloads/, player/, monsters/, ui/, abilities/.	T
LPROJ-003	CORE-001	Autoloads shall include: DataLoader, SaveManager, InputManager, TimeManager, SeasonManager, WeatherManager, AreaManager, HexGrid, SettlementManager, SpawnManager, DisciplineManager, AudioManager, MusicManager, TradeGraph, FoundingManager, PlayerStats.	T
LPROJ-004	CORE-005	Data directory: res://data/ containing all JSON files.	T

LLR	Traces To	Requirement	T/U
LPROJ- 005	CORE- 004	User data: user://saves/ for save files. user://input_config.json for custom keybindings.	T

End of Document